数学物理与几何研讨会会议手册

数学物理与几何是现代数学的核心分支之一。本次会议旨在邀请数学物理与几何领域的专家学者一起研讨该领域的重要进展，增进同行之间的学术交流与合作，并为国内的青年学者和学生了解该方向提供平台。我们诚挚地欢迎您的参与！

本会议不收取会议注册费。会议的其他事项通知如下：

一. **邀请报告人(按姓氏首字母顺序排列)**:

戴波　　（北京大学）
方博汉　　（北京大学）
郭帅　　（北京大学）
何伟强　　（中山大学）
柯华忠　　（中山大学）
李长征　　（中山大学）
李勤　　（南方科技大学）
Ryszard Nest　(University of Copenhagen & BIMSA）
孙善忠　　（首都师范大学）
涂君武　　（上海科技大学）
王路瑶　　（北京雁栖湖应用数学研究院）
王新　　（山东大学）
邬龙挺　　（南方科技大学）
吴朝中　　（中山大学）
杨迪　　（中国科学技术大学）
周春辉　　（中国科学技术大学）
周坚　　（清华大学）
宗正宇　　（清华大学）

二. 会议时间： 2026 年 4 月 17 日-4 月 19 日，4 月 16 日下午报到。

三. 会议地点： 华中科技大学逸夫科技楼南楼 702

四. 主办单位： 华中科技大学数学与统计学院

华中科技大学数学中心

五. 会议联系人:

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会议日程

时间 日期	4 月 17 日	4 月 18 日	4 月 19 日
8:40 - 8:45	**开幕**		
8:45 - 9:25	**戴波**	**周坚**	**孙善忠**
9:30 - 10:10	**李长征**	**方博汉**	**宗正宇**
10:10 - 10:40	**茶歇**		
10:40 - 11:20	**郭帅**	**涂君武**	**Ryszard Nest**
11:25 - 12:05	**杨迪**	**吴朝中**	**邬龙挺**
12:10 - 14:30	**午餐（绿园自助餐厅）**		
14:30 - 15:10	**何伟强**	**李勤**	**自由讨论，离会**
15:15 - 15:55	**周春辉**	**王路瑶**	
15:55 - 16:25	**茶歇**	**自由讨论**	
16:25 - 17:05	**王新**		
17:10 - 17:50	**柯华忠**		
18:00 - 20:00	**晚餐（绿园自助餐厅）**	**晚宴(湖锦酒楼)**	

题目和摘要(按报告人姓氏首字母顺序排列)

戴波（北京大学）

题目: Soliton solutions of ASDYM equation on $\mathbb{R}^{2,2}$ and its reductions
摘要: Many integrable system can be reduced from the anti-self-dual Yang-Mills (ASDYM) equations on $\mathbb{R}^{2,2}$. I will explain a method developed by Chuu-Lian Terng and myself, which can be used to construct all soliton solutions to the ASDYM equation on $\mathbb{R}^{2,2}$ with structure group U(n) or SU(n), and some of its reductions, such as spacetime monopole equation, modified 2+1 chiral model, 2d chiral model, and others. This method is based on loop group factorization, include a few versions of Backlund transformations and a limiting method.

方博汉（北京大学）

题目: Conic bundle and the mirror hypersurface for a toric Calabi-Yau n-fold
摘要: We study the Hori-Vafa mirror Calabi-Yau n-fold to a toric Calabi-Yau n-orbifold. One version of homological mirror symmetry expects that the wrapped Fukaya category on the Hori-Vafa mirror is equivalent to the toric Calabi-Yau with a hypersurface removed. We use a microlocal sheaf model of this Fukaya category, and in this setting prove such statement by descent property and Orlov's semi-orthogonal decomposition. Moreover, we expect the object-level correspondence when interpreted in terms of some reasonably defined characteristic cycles, matches the integral structure correspondence from the Gamma classes. This talk is based on the joint work with Yuze Sun and Peng Zhou.

郭帅（北京大学）

题目: Structure of higher-genus Gromov-Witten Theory of $\mathcal{O}(p-1) \oplus \mathcal{O}(-p-1)$
摘要: We consider the Gromov-Witten theory of the total space of the rank 2 bundle $\mathcal{O}(p-1) \oplus \mathcal{O}(-p-1)$ over $\mathbb{P}^1$. Our aim is to prove a structural result conjectured by Caporaso, Griguolo, Mariño, Pasquetti, and Seminara. Moreover, for the case p≥2, we try to construct and prove the spectral curve for these models, which is not covered by the Remodeling Conjecture proved by Fang–Liu–Zong. This is based on joint work with Jingyi Xu and Qingsheng Zhang.

何伟强（中山大学）

题目: Landau-Ginzburg mirror symmetry for $x^p + y^q$

摘要: Suppose W is a quasi-homogenous invertible polynomial and G is some admissible symmetry groups of W. The Landau-Ginzburg (LG) mirror symmetry conjecture states the equivalence between LG A model on the pair (W, G) and LG B model on the mirror pair (W^T, G^T). Here LG A model is induced by curve counting and LG B model is induced by variation of Hodge structure. This talk includes an introduction on LG A B model and a recent progress for the proof of the above conjecture in the case $x^p + y^q$, this is a joint work with Si Li and Yifan Li.

柯华忠（中山大学）

题目: Gamma conjecture II for del Pezzo surfaces via Gamma-I

摘要: For quantum cohomology of a Fano manifold X, Gamma conjecture I and II try to describe the asymptotic behavior of Dubrovin connection in terms of derived category of coherent sheaves on X, via the Gamma-integral structure. Gamma conjecture I relates the structure sheaf to a flat section with the smallest asymptotics, and Gamma conjecture II assumes semisimplicity and relates full exceptional collections to asymptotic exponential fundamental solutions. In this talk, I will propose a strategy to verify Gamma conjecture II via Gamma-I, and apply this strategy to the case of del Pezzo surfaces. This talk is based on joint work with Jianxun Hu, Changzheng Li and Zhitong Su.

李长征（中山大学）

题目: Quantum Schubert calculus for smooth Schubert divisors in Fl_n

摘要: The cup product of Schubert classes in the classical cohomology of Schubert varieties can be directly read off from that in the classical cohomology of flag varieties. Unlike ordinary cohomology, quantum Schubert calculus for Schubert varieties does not follow from that of flag varieties due to lack of functorial properties in quantum cohomology. In this talk, we will discuss the quantum Schubert calculus for smooth Schubert divisors X in the complete flag varieties Fl_n, including a Borel-type ring presentation of the quantum cohomology, the quantum Schubert polynomials and the Monk-Chevalley formula. We will also explain how this was motived by an on-going project on Peterson program initiated by Rietsch, Yang and myself. This talk is based on my joint work with Jiayu Song, Rui Xiong and Mingzhi Yang.

李勤（南方科技大学）

题目: Quantization of Kahler manifolds via differential operators

摘要: Berezin-Toeplitz (B-T) operators are one of the most important methods to construct quantization on Kahler and symplectic manifolds. For every function, the associated B-T operators are defined as the composition of the multiplication and the orthogonal projection on the Hilbert spaces. In this talk, I will describe a sequence of differential operators which generalize the classical Kostant-Souriau operators, and show that the BT operators are asymptotic to these operators. This implies the asymptotic locality of B-T operators. This is a joint work with Kwokwai Chan, Conan Leung and Yutung Yau.

Ryszard Nest（University of Copenhagen & BIMSA）

Title: A secondary index for chirality operators associated with quantum walks.

Abstract: The chirality index is a K-theoretic invariant associated to a couple of almost anti-commuting symmetries on a Hilbert space. It appears in physics is connected for example quantum random on lattices. We will recall the construction of random quantum walk on graphs, the as sociated chiral operators and the computation of the associated K-theoretic invariant in a few examples, including the case of quantum rand om walk on a binary tree. This is a joint work with Toshikazu Natsume.

孙善忠（首都师范大学）

题目：Around Witten's Asymptotic Expansion Conjecture

摘要：Gukov-Pei-Putrov-Vafa (GPPV) invariant is a refinement of the well-known Witten-Reshetikhin-Turaev (WRT) invariant in quantum topology. I will talk about our recent progress on the quantum modularity of GPPV invariants for Seifert fibered integral homology spheres (SFIHS) using resurgence theory, and its applications to the asymptotic expansion conjecture through the non-tangential limit to the WRT invariants. Its relation to Habiro invariant will also be addressed if time permitted. Based on joint works with J. Andersen, L. Han, Y. Li, W. Mistergard and D. Sauzin.

涂君武（上海科技大学）

题目：B-model categorical enumerative invariants and the holomorphic anomaly equation

摘要：In this talk we consider categorical enumerative invariants associated with derived category of coherent sheaves on smooth projective Calabi-Yau threefolds. We prove these invariants, when considered in a family, satisfy Bershadsky-Cecotti-Ooguri-Vafa's holomorphic anomaly equation (HAE). We also discuss the implications of HAE to understand the geometry of moduli space of Calabi-Yau threefolds.

王路瑶（北京雁栖湖应用数学研究院）

题目: Geometric realization of W-operators

摘要: Starting from the transposition class sum in the center of the symmetric group algebra and the classical cut-and-join operator, we construct cubic W-operators on symmetric functions via the characteristic map. Using the Grojnowski–Nakajima Fock space realization, these algebraic operators are then mapped to the equivariant cohomology of $Hilb^n(\mathbb{C}^2)$. In this picture, the raising operator E_1 is realized by the Hecke correspondence, while the β-deformed cubic generator $W_0^{(\beta)}$ is interpreted as a combination of triple incidence correspondences and diagonal correction terms. In this talk, I will also briefly discuss the fixed-point localization viewpoint on matrix elements and possible extensions to quiver varieties, (q,t)-deformations, and refined Hurwitz theory. This talk is based on my recent work: Geometric realization of W-operators (arXiv:2602.04779).

王新（山东大学）

题目: Tautological Relations on the Moduli Space of Curves

摘要: In the first part, we review several formulae for the Hodge character class $\chi_{2g-1}(E)$ on $\overline{\mathcal{M}_g}$, Liu–Pandharipande relations and the Blot–Shadrin–Singh relations, and discuss their consequences. In the second part, we construct new explicit relations among Hodge classes, psi-classes, and boundary classes. Using these results, we derive new universal equations for descendant Hodge integrals.

邬龙挺（南方科技大学）

题目: Poincaré polynomials of moduli spaces of 1-dimensional sheaves on the projective plane

摘要: The geometry of moduli spaces of one-dimensional sheaves on the projective plane has attracted a lot of study recently. In this talk, I will give a new calculation of the Betti numbers of the moduli spaces of one-dimensional sheaves on the projective plane using Gromov-Witten invariants of local $\mathbb{P}^2$ and local curves. The new calculation is based on the refined sheaves/GW correspondence established by Bousseau and all genus local/relative correspondence given by Bousseau-Fan-Guo-Wu. It can be used to prove the divisibility property of Poincaré polynomials of moduli spaces of one-dimensional sheaves on projective plane conjectured by Choi-van Garrel-Katz-Takahashi, and can also be used to determine the leading Betti numbers. Some conjectures concerning the higher range Betti numbers will be proposed if time permits. This is based on a joint work with Shuai Guo and Miguel Moreira.

吴朝中（中山大学）

题目: Additional symmetries of the KP-mKP hierarchy and Virasoro constraints to the Burgers-KdV hierarchy

摘要: For the KP-mKP hierarchy introduced via pseudo-differential operators containing two derivations, we derive a class of (differential) Fay identities and construct a series of additional symmetries. Moreover, the additional symmetries are represented as certain linear actions on the tau functions of the hierarchy, which lead to a new proof of the Virasoro constraints to the tau functions of the Burgers-KdV hierarchy. This work is collaborated with Zongyao Feng and Lumin Geng.

杨迪（中国科学技术大学）

题目: Uniform large genus asymptotics for Br\'ezin--Gross--Witten numbers and Witten's intersection numbers

摘要: Br\'ezin--Gross--Witten numbers and Witten's intersection numbers are two families of rational numbers, which have physical origins and topological meanings. Their partition functions are tau-functions of the KdV hierarchy. In this talk, we study arithmetic and combinatorial properties of these numbers, and give their uniform large genus asymptotics. The talk is based on joint works with Jindong Guo, Paul Norbury and Don Zagier (see arXiv:2412.20388, arXiv:2603.15233).

周春辉（中国科学技术大学）

题目：On the Burgers-KdV hierarchy and its tau-function

摘要：In this talk, we will discuss about the Burgers-KdV hierarchy which is constructed by A. Buryak to describe the so-called open intersection numbers. This hierarchy can be seen as a kind of extension of KdV hierarchy and its tau-function is also closely related to the KdV's tau-function. We will show how to use Sato theory and Dubrovin-Zhang theory to derive some properties of Burgers-KdV hierarchy and show some applications on the related quasi-trivial transformation.

周坚（清华大学）

题目：From probability theory to Gromov-Witten theory, and back

摘要：We introduce a notion of extended matrix models, inspired by Gromov-Witten theory of the projective line. This establishes a link between the space of one-dimensional probability distribution to the big phase space of the GW theory of the projective line. In this perspective, we consider the probability distribution corresponding to the GW theory of the projective line.

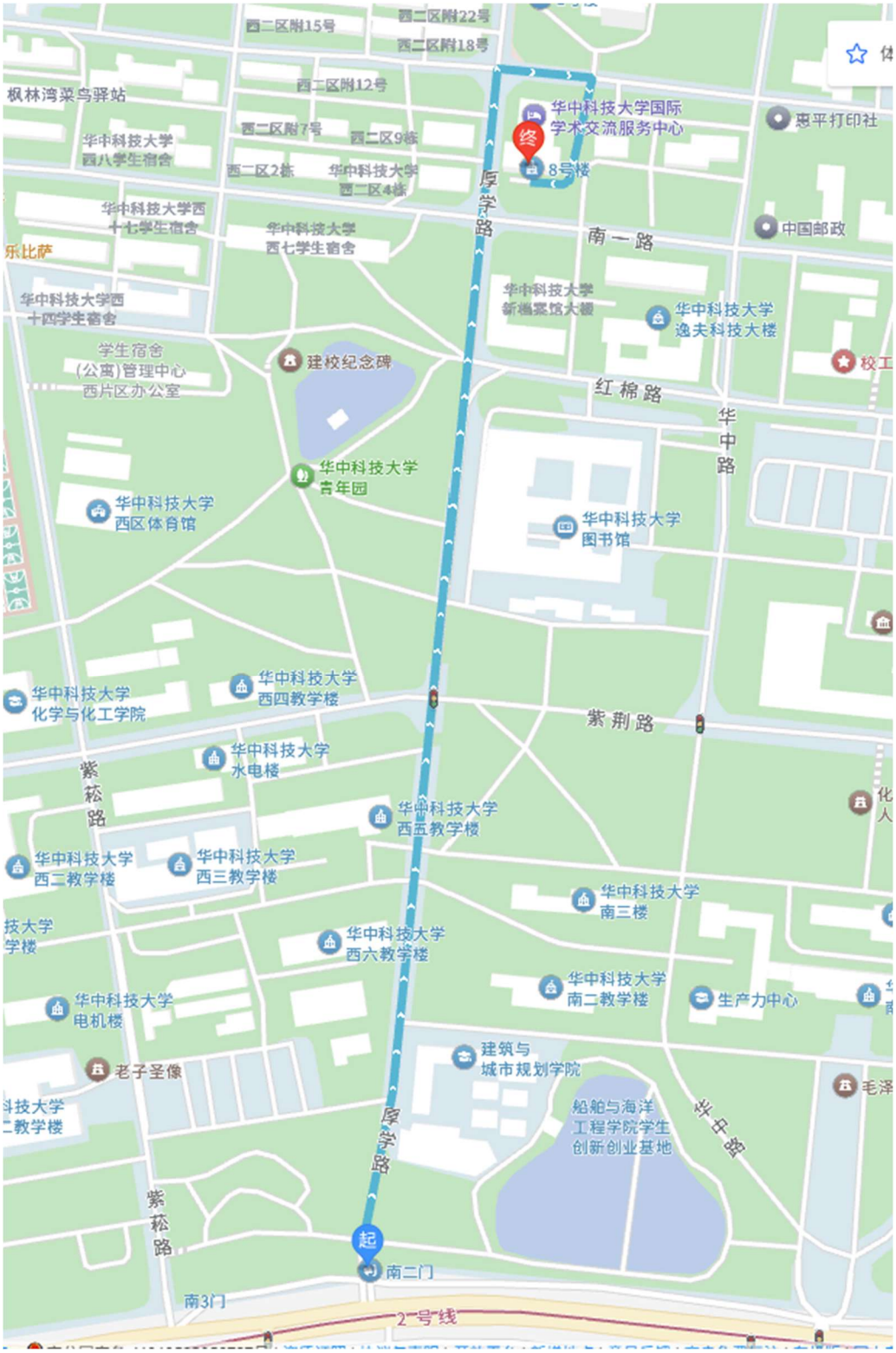
宗正宇（清华大学）

题目: All genus Crepant Transformation Conjecture for toric Calabi-Yau 3-orbifolds

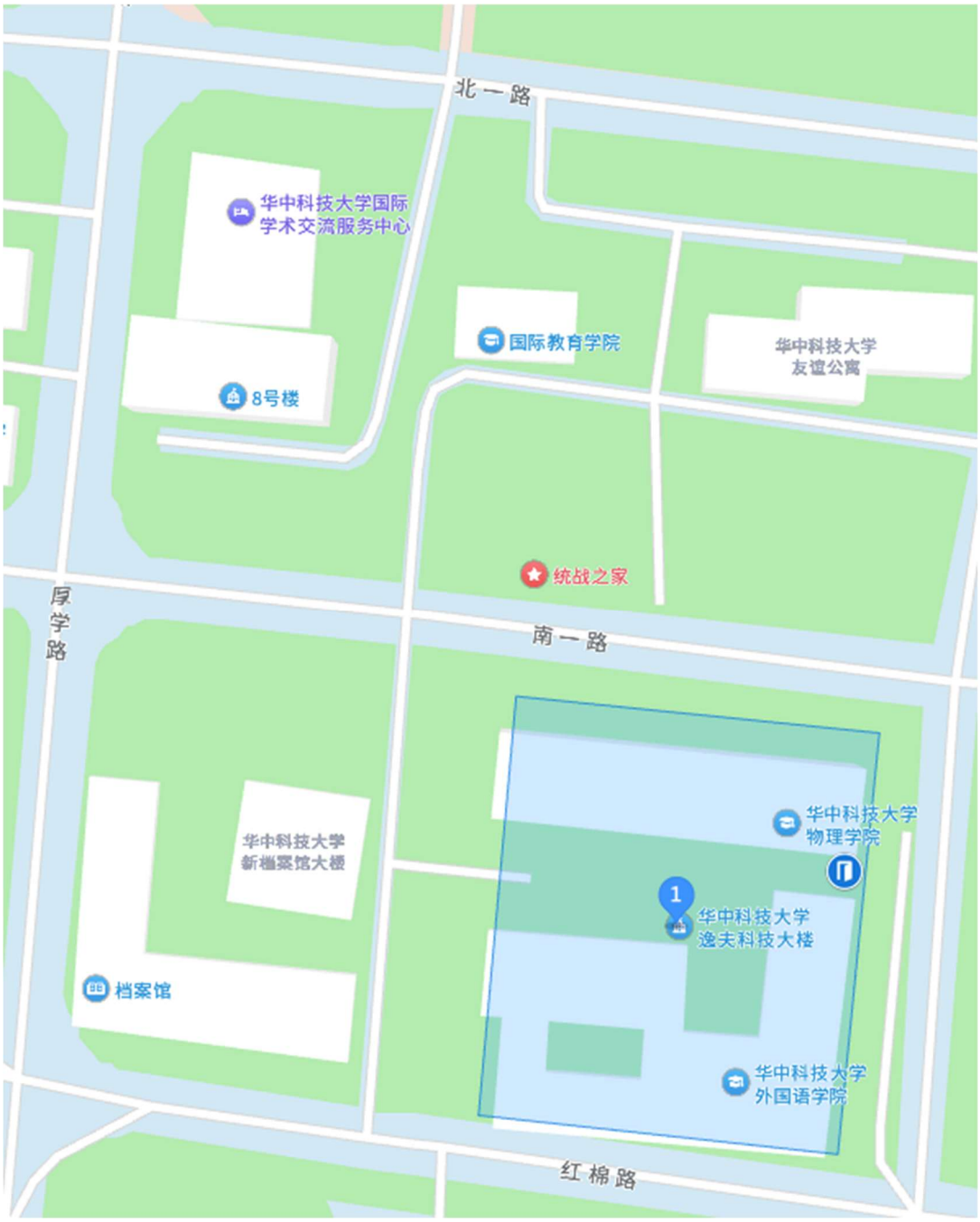
摘要: The Crepant Transformation Conjecture, proposed by Ruan, asserts certain equivalence between Gromov-Witten theory of two manifolds/orbifolds which are related by a crepant transformation. In general, the higher genus Crepant Transformation Conjecture is quite hard to study. Even the formulation of the Crepant Transformation Conjecture in the higher genus case is subtle. In this talk, I will explain the proof of the all genus Crepant Transformation Conjecture for general toric Calabi-Yau 3-orbifolds. We will consider the all genus open, closed, descendant Gromov-Witten theory of toric Calabi-Yau 3-orbifolds. This talk is based on a project joint with Bohan Fang, Chiu-Chu Melissa Liu, and Song Yu.

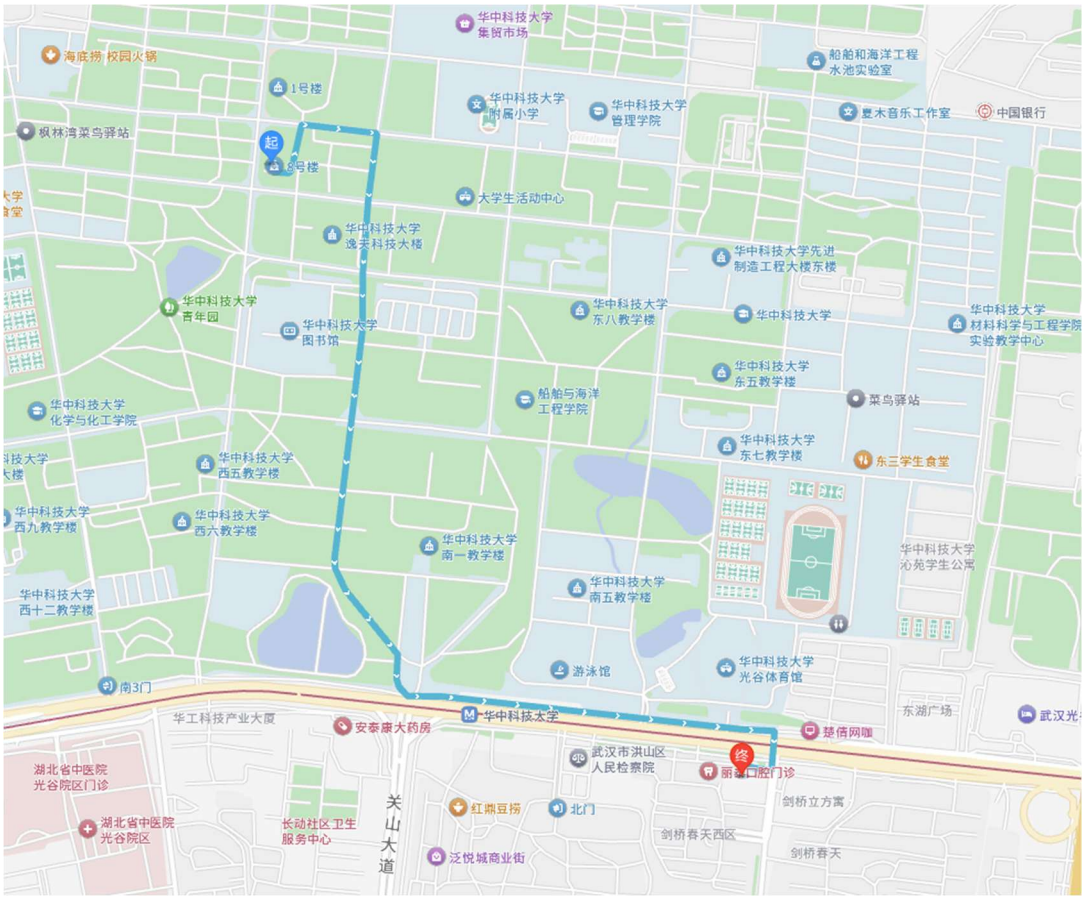
住宿地点：华中科技大学国际学术交流服务中心 8 号楼。
交通：
1. 从武汉火车站坐**地铁 4 号线到中南路站转地铁 2 号线**（约 50 分钟），到**华中科技大学站**，D 出口出站，经**南大门**入校后步行约 15 分钟至酒店。
2. 从天河机场可以直接坐**地铁 2 号线**到**华中科技大学站**（约 90 分钟），D 出口出站，经**南大门**入校后步行约 15 分钟至酒店。

3. 乘坐出租车可选择华中科技大学（主校区）南二门下车，经南二门入校后向北步行约 10 分钟至酒店。（不堵车的情况下，天河机场打车至南二门约 50 分钟；武汉站打车至南二门约 25 分钟。如乘坐打表出租车，可凭纸质小票报销。）



会议地点：华中科技大学逸夫科技楼南楼 702 会议室。从酒店 8 号楼出发，向东南方向步行约 2 分钟到达逸夫科技楼。（逸夫科技楼分为南楼和北楼，请步行至南侧睿言空间咖啡店旁的电梯乘坐至 7 楼。）



午餐及晚餐地点（4 月 16 日晚餐，4 月 17 日午餐及晚餐，4 月 18 日与 19 日午餐）：华中科技大学绿园自助餐厅。从酒店 8 号楼出发，向北步行约 2 分钟到达绿园餐厅，凭会议发放的餐券就餐。

4 月 18 日晚宴地点：湖锦酒楼光谷店。从酒店 8 号楼出发步行约 25 分钟到达。